



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-311-1

Date: 20-9-2022

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SPECIFICATION

FOR

Protective Fence specification

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CONTENTS

<u>Clause</u>	<u>Description</u>	<u>Page</u>
1	Scope	3
2	Required Specification	3
3	Installation	4



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1. SCOPE

This specification covers the supply of a protective fence on distribution transformers of medium voltage for the purpose of setting up a protection zone from penetration of passing people, especially children, to protect them from shock by electric current ,

The specification covers supply (including fabrication) Providing & fixing of Chain link fencing using MS pipe ,MS angle & GI wire Mesh chain link for distribution Transformer centers/RMUs as per design & drawing of UGVCL.

For the safety aspects, to cover distribution Transformer Centre/RMU , a chain link fencing is to be fabricated from MS pipe of square hollow section(SHS) ,MS angle, MS Flat & GI wire as per the design & drawing and to be provided/fitted with vertical support posts of MS pipe of Square Hollow Section (SHS) in the ground using standard cement concrete mixture.

2. REQUIRED SPECIFICATION:-

The protective fence consists of:

- i. The main beam (consisting of a hollow square iron size 50 mm X 50 mm X 5 mm with an approximate weight of 6 kg / linear meter)
- ii. The outer frame beam (consisting of a hollow square iron, size 32 mm X 32 mm X 3.2 mm, with an approximate weight of 2.7 kg / linear meter).
- iii. The frame beam on which the net is installed (an angle iron beam of size 25mm X 25mm X 3 mm with an approximate weight of 1.1 kg / linear meter).
- iv. Wire mesh (consisting of a square mesh 2 inches x 2 inches thickness of 5.25 mm² - greased 40 microns, minimum limit of zinc
- v. To connect the main beam and the outer frame with a flat beam (wicker iron) 50 mm X 5 mm .



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- vi. Good quality and durable bolt hinges are used and are well fixed for good welding.

3. INSTALLATION:

- i. The main columns (the main beam) are installed, measuring 50mm X 50mm X 5 mm with a length of not less than 2.1 meters with 5 columns with dimensions according to the attached drawing and they are fixed to the ground through a concrete base with a homogeneous mixture of the usual concrete mix (300kg cement , 0.8 m³ asphalt , 0.4 m³ sand per cubic meter) and the dimensions of the concrete base are 30 cm x 30 cm x 50 cm.
- ii. The outer frame is installed 32mm X 32mm X 3.2 mm according to the dimensions shown in the attached engineering drawing and is fixed to the main columns using wicker iron size 50 mm X 5 mm with good welding.
- iii. The frame on which the mesh is installed is an angle iron beam of size 25mm X 25mm X 3 mm by good welding in the outer frame.
- iv. The wire mesh is installed (which is a square mesh 2 inches X 2 inches thickness of 5.25 mm 2 - greased 40 microns minimum of zinc by good welding in the corner frame.
- v. Hinges and bolts are installed in suitable sizes with good welding according to the attached engineering drawing.



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- xiv. The quality of the finishing must be ensured by the responsible engineer.
- xxiii. The fence is painted with a layer of primer and two layers of oil.
- xxxiv. The supervising engineer must ensure the dimensions of the installation, the quality of the welds, and the new installation in the ground according to the specifications.
- xliv. The height of the fence according to the engineering drawing is 1.5 meters from the surface of the earth, and the height can be adjusted according to the location on the ground with an adjustment of the height (no more than 2.5 meters from the surface of the earth) and an increase in the section of the concrete base so that it is 30 cm X 30 cm X 70 cm..



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Fencing Drawing

Transformer Center Fencing View

